# Lesson 12 Teacher's Key: Hearing It Happen

Scratch - Instructor Reference Only

VIDEO GAME CREATION

## What This Lesson Teaches

Formalizing earlier lessons' optional 'add a sound' ideas (Lessons 3, 4, 6, 8) into mandatory sounds, plus one genuinely new one (wave clear), each placed in the exact script Rule 13 already says owns that decision - no new Scratch concept, just recognizing where an event already happens and adding one block there.

## Level 1 - Full Worked Solution

- sfx_hit moves onto the Enemy sprite's own touching-Bullet branch, right where it already broadcasts [enemy hit] - Enemy still never touches score or enemies_remaining itself (Rule 13); it only reports that a hit happened.
- sfx_wave is the one genuinely new sound this week, added to GameManager's existing "when I receive [enemy hit]" script, in the else branch Lesson 11 already built.
- sfx_gameover (from Lesson 8's optional idea, not a new asset) becomes mandatory today and moves to the Player sprite's "when I receive [game over]" script, where Lesson 8 placed it - if a student already built it, nothing changes; if not, this is the one line of 'old' optional content this lesson actually requires.
- If a student built Lesson 6's optional sfx_win, it needs zero changes - it already lives in the correct (true-win) branch of the same nested if on GameManager.

## Why This Structure

- Every sound this week lives inside a script that only runs once per actual event (a hit, a wave clear, a loss, a game over) - Scratch's clone-and-broadcast model has made every trigger point naturally one-shot since Lesson 5, so there's no risk of a sound replaying every frame.
- Keeping sfx_wave and sfx_win in opposite branches of the same if/else on GameManager (rather than, say, always playing sfx_wave and separately checking whether to also play sfx_win) guarantees they can never both fire on the same event - a direct benefit of Lesson 11's branching structure.
- sfx_hit staying on Enemy and sfx_wave/sfx_win staying on GameManager is the same Rule 13 split this project has kept since Lesson 6: Enemy reports, GameManager decides. There is no reason for this lesson's sounds to be the first thing that breaks that pattern.

## Level 2 / Level 3 Talking Points

- Ask students who already had sfx_shoot/sfx_hit/sfx_win/sfx_gameover from earlier optional lessons to identify, out loud, exactly which one sound is genuinely new to them today (sfx_wave) versus which one just became mandatory (sfx_gameover) - a good way to surface who's starting fresh versus who's just adding one line.
- For students ready for a stretch: ask what would happen if sfx_hit were moved from Enemy's touching-Bullet branch onto GameManager's "when I receive [enemy hit]" script instead - it would still work (both run once per hit), but Enemy's sprite would feel less 'alive' on its own. A good prompt for noticing that correctness and feel aren't always the same question.

## Common Student Mistakes

- Placing sfx_wave or sfx_win before the if/else split on GameManager instead of inside their respective branches, causing both to play on every wave clear including the true win.
- Uploading a new sound file to the wrong sprite (Scratch sounds are per-sprite) and wondering why "start sound" doesn't find it.
- Forgetting that a student who skipped Lessons 3/4/6/8 entirely needs full instructions this week, not just 'add one sound' - this lesson's Handout is written to work for both cases.